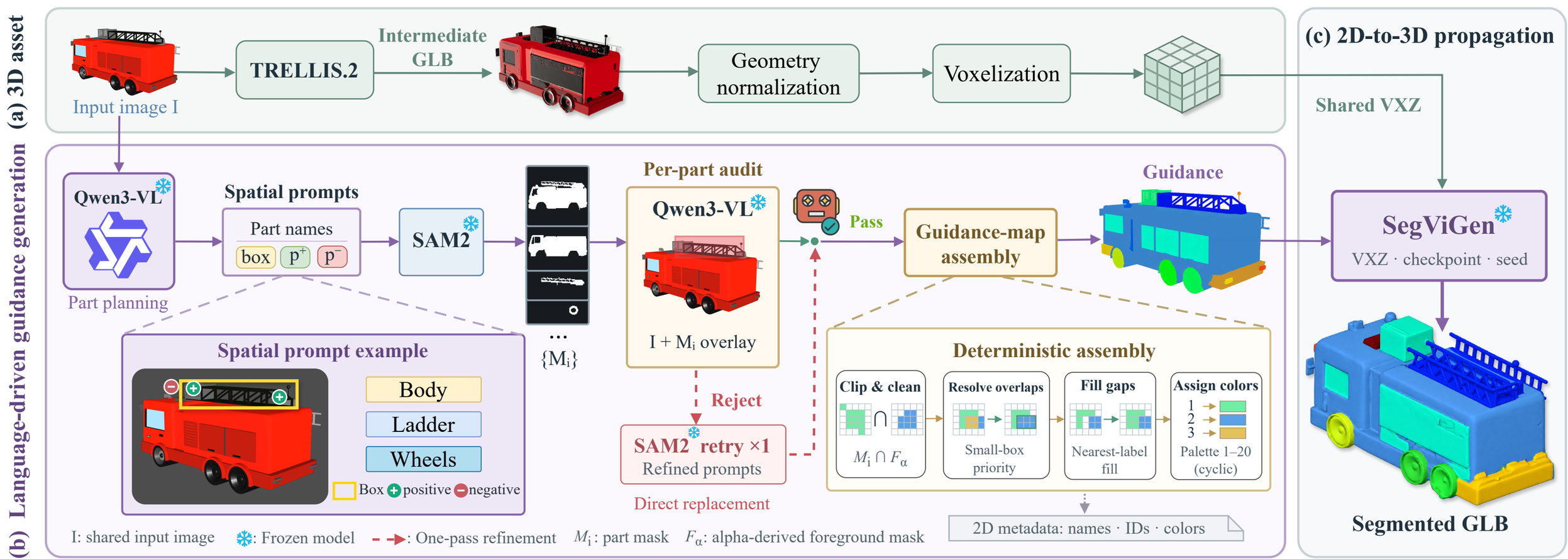

(a) 3D asset
Input image I
TRELLIS.2
Intermediate GLB
Geometry normalization
Voxelization
(c) 2D-to-3D propagation
Shared VXZ
(b) Language-driven guidance generation
Qwen3-VL
Part planning
Spatial prompts
Part names
box
p+
p−
SAM2
{Mi}
Per-part audit
Qwen3-VL
I + Mi overlay
Pass
Reject
SAM2 retry ×1
Refined prompts
Direct replacement
Guidance
Guidance-map assembly
Deterministic assembly
Clip & clean
Mi ∩ Fα
Resolve overlaps
Small-box priority
Fill gaps
Nearest-label fill
Assign colors
1
2
3
Palette 1–20 (cyclic)
2D metadata: names · IDs · colors
Spatial prompt example
Body
Ladder
Wheels
Box positive negative
SegViGen
VXZ · checkpoint · seed
Segmented GLB
I: shared input image
: Frozen model
: One-pass refinement
Mi : part mask
Fα: alpha-derived foreground mask